

IVALO, Finland

WMO: 028070

Lat: 68.608N

Lon: 27.414E

Elev: 481

StdP: 14.44

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.9	-20.1	-32.6	0.9	-26.7	-25.4	1.4	-19.9	22.1	22.8	19.5	23.8	2.7	240	0.727

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	76.7	61.1	72.4	59.1	68.5	57.6	63.5	72.5	61.2	69.3	59.0	66.4	8.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.8	78.1	66.6	57.5	71.9	64.2	55.6	66.9	62.1	29.1	72.1	27.4	69.6	25.9	66.4	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.8	16.4	14.4	-32.1	82.0	7.2	3.9	-37.3	84.8	-41.6	87.1	-45.6	89.3	-50.9	92.1		
			-32.2	65.9	6.9	2.6	-37.2	67.7	-41.2	69.2	-45.1	70.7	-50.1	72.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	32.0	9.7	10.2	18.5	29.3	40.2	50.9	57.6	53.3	44.6	32.5	20.7	14.6
	DBStd	19.39	14.33	14.55	11.01	8.52	7.59	7.51	6.85	6.38	6.55	8.37	11.26	13.58
	HDD50	7112	1250	1114	978	621	320	78	12	35	187	542	879	1096
	HDD65	12080	1715	1534	1443	1071	769	428	245	366	613	1006	1329	1561
	CDD50	531	0	0	0	0	16	104	249	138	24	0	0	0
	CDD65	25	0	0	0	0	0	4	16	5	0	0	0	0
	CDH74	242	0	0	0	0	7	39	151	45	0	0	0	0
	CDH80	33	0	0	0	0	0	5	22	6	0	0	0	0
Wind	WSAvg	6.2	5.6	6.1	6.7	6.3	6.9	7.0	6.3	5.5	5.9	6.1	5.7	6.2
Precipitation	PrecAvg	15.5	0.8	0.6	0.7	0.7	1.1	1.7	2.5	2.4	1.7	1.4	1.0	0.8
	PrecMax	20.5	1.6	2.0	2.4	1.6	2.4	3.5	4.2	5.0	3.5	4.3	2.4	1.5
	PrecMin	10.9	0.1	0.2	0.1	0.2	0.1	0.2	0.6	0.5	0.3	0.4	0.4	0.3
	PrecStd	2.6	0.4	0.4	0.5	0.4	0.6	0.9	0.9	1.1	0.8	0.8	0.5	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.0	38.5	42.6	54.3	72.5	79.2	83.0	78.9	65.1	52.4	42.5	39.3
		MCWB	34.0	35.2	36.5	44.5	55.1	61.0	63.5	63.6	56.2	48.8	41.0	36.3
	2%	DB	33.1	34.3	37.7	48.0	64.3	72.5	78.2	73.7	61.2	49.3	37.8	35.8
		MCWB	31.7	32.2	33.6	40.3	50.7	57.0	61.6	60.9	53.7	46.1	36.4	34.0
	5%	DB	30.3	31.9	35.1	44.2	57.8	67.9	74.1	68.7	58.1	46.5	35.7	33.8
		MCWB	28.9	30.2	31.8	38.0	47.4	54.9	60.4	58.9	52.4	44.2	34.5	32.4
	10%	DB	27.3	29.1	32.5	41.0	53.2	63.8	70.2	64.6	55.2	43.5	34.0	31.6
		MCWB	26.2	27.6	29.8	35.6	45.2	53.1	59.1	57.0	50.9	41.5	33.0	30.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.4	35.8	36.9	45.2	55.8	63.2	66.5	65.6	57.6	49.7	40.9	36.4
		MCDB	36.1	38.5	41.7	53.3	69.2	73.4	75.9	75.9	63.1	51.9	42.3	38.3
	2%	WB	31.7	32.4	34.2	41.0	52.1	59.4	64.4	62.5	55.2	46.5	36.6	34.2
		MCDB	32.8	34.2	37.2	46.6	62.4	69.5	73.8	71.1	59.6	48.4	37.5	35.6
	5%	WB	28.8	30.2	32.0	38.4	48.4	56.6	62.3	60.0	53.0	44.2	34.6	32.5
		MCDB	30.1	31.8	34.5	43.5	56.9	64.5	71.0	66.6	57.2	46.4	35.4	33.8
	10%	WB	26.1	27.6	30.0	36.3	45.7	54.4	60.3	57.7	51.2	41.5	33.0	30.0
		MCDB	27.3	29.1	32.6	40.0	52.2	62.7	68.0	63.7	54.8	43.4	33.9	31.4
Mean Daily Temperature Range	5% DB	MDBR	14.1	14.6	17.4	16.0	14.3	14.3	14.8	14.6	12.5	8.8	11.0	12.9
		MCDBR	12.2	9.9	13.9	17.4	23.0	21.4	21.8	21.6	17.1	9.7	8.2	10.2
		MCWBR	11.4	8.8	11.1	11.2	12.3	9.5	9.4	10.7	10.7	7.9	7.7	9.4
	5% WB	MCDBR	12.3	10.1	12.2	15.8	21.3	18.3	18.6	18.8	14.6	9.1	7.4	10.3
		MCWBR	11.6	9.1	10.2	10.5	12.4	9.4	9.4	10.2	9.8	7.7	7.1	9.6
Clear-Sky Solar Irradiance	taub	0.194	0.248	0.290	0.321	0.342	0.343	0.355	0.337	0.306	0.267	0.354	N/A	
	taud	1.917	2.069	2.153	2.219	2.406	2.477	2.465	2.492	2.547	2.284	2.732	N/A	
	Ebn at Noon	20	165	231	259	269	272	264	256	237	163	35	0	
	Edh at Noon	2	15	25	32	30	29	29	25	18	12	3	0	
All-Sky Solar Radiation	RadAvg	9	147	616	1195	1443	1541	1409	1001	528	169	24	0	
	RadStd	1	13	36	70	149	134	157	93	49	27	3	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+1.08	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	+0.99	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page